1️⃣ Identify and describe the mathematical aspects of a real-world problem situation (Most Relevant)

Relevant item:
6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables formulate

How it is reflected

Students must extract key mathematical information from the real-life situation.

Key variables include:

- Number of DVDs x
- Cost for non-members: $3.20x$
- Cost for members: $10 + 2.50x$

Students must recognize that:

- The membership fee is a **fixed cost**.
- The rental fee is a **variable cost** depending on the number of DVDs.

Mathematical representation

Non-member cost

$$C_1 = 3.20x$$

Member cost

$$C_2 = 10 + 2.50x$$

Then the two costs must be compared.

Why it is the most relevant

The essence of the problem is **translating a real-world situation into mathematical variables and relationships**.

2️⃣ Identify the key variables in a model

Relevant item:
2. Identify the key variables in a model formulate

How it is reflected

Students must determine:

Core variable:

- x: number of DVDs

Related parameters:

- 3.20 (non-member price per DVD)
- 2.50 (member price per DVD)
- 10 (membership fee)

Without identifying the variable, the model cannot be established.

Ability reflected

This represents the **ability to identify variables in mathematical modelling**.

3️⃣ Translate a problem into a standard mathematical representation

Relevant item:

9. Translate a problem into a standard mathematical representation or algorithm formulate

How it is reflected

Students must translate the verbal problem into a mathematical inequality:

$$10 + 2.50x \leq 3.20x$$

which means:

Member cost ≤ Non-member cost.

Solving the inequality:

$$10 \leq 0.70x$$

$$x \geq 14.29$$

Therefore:

The minimum number of DVDs is 15.

Ability reflected

This step demonstrates the ability to convert a **real-world situation into a mathematical inequality model**.

4️⃣ Select a mathematical description or representation

Relevant item:

1. Select a mathematical description or representation that describes a problem formulate

How it is reflected

Students must select an appropriate mathematical representation, such as:

- Algebraic expressions
- Inequality models
- Cost functions

The most natural representation in this problem is **a comparison of linear cost functions**.

5️⃣ Recognise mathematical structure (relationships and patterns)

Relevant item:

5. Recognise mathematical structure (including regularities, relationships, and patterns) formulate

How it is reflected

Students must recognize that this problem involves a **linear relationship**:

Cost structure:

$$C = a + bx$$

where

- a = fixed cost
- b = cost per unit

This is a typical **cost function model in real-world contexts**.

6️⃣ Identify constraints, assumptions, and simplifications

Relevant item:

12. Identify the constraints, assumptions simplifications in a mathematical model formulate

How it is reflected

Implicit modelling assumptions include:

- The price per DVD is constant.
- No discounts are considered.
- The number of DVDs must be an integer.
- The membership fee is paid once per year.

These represent **simplifications in the modelling process**.

1️⃣ Identify and describe the mathematical aspects of a real-world problem situation（最相关）

对应条目：

6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables formulate

体现方式

学生需要从现实情境中提取关键数学信息：

关键变量包括：

- DVD 数量 x
- 非会员费用 $3.20x$
- 会员费用 $10 + 2.50x$

学生必须理解：

- 年费是固定成本
- 租金是随数量变化的变量成本

数学化表达

建立费用函数：

非会员费用

$$C_1 = 3.20x$$

会员费用

$$C_2 = 10 + 2.50x$$

再比较两者。

为什么最相关

因为题目本质就是**将生活情境转化为数学变量与关系。**

2️⃣ Identify the key variables in a model

对应条目：

2. Identify the key variables in a model formulate

体现方式

学生必须确定：

核心变量：

- x：DVD数量

相关参数：

- 3.20（非会员单价）
- 2.50（会员单价）
- 10（会员费）

如果不能识别变量，就无法建立模型。

体现能力

这是建模中的**变量识别能力。**

3️⃣ Translate a problem into a standard mathematical representation

对应条目：

9. Translate a problem into a standard mathematical representation or algorithm formulate

体现方式

学生需要把语言问题转换为数学不等式：

$$10 + 2.50x \leq 3.20x$$

表示：

会员费用 ≤ 非会员费用

然后求解：

$$10 \leq 0.70x$$

$$x \geq 14.29$$

因此：

最少需要 15 张 DVD

体现能力

这是典型的：

现实情境 → 数学不等式模型

4️⃣ Select a mathematical description or representation

对应条目：

1. Select a mathematical description or representation that describes a problem formulate

体现方式

学生需要选择合适的数学表示方式，例如：

- 代数表达式
- 不等式模型
- 成本函数

本题最自然的表示是：

线性成本函数比较。

5️⃣ Recognise mathematical structure (relationships and patterns)

对应条目：

5. Recognise mathematical structure (including regularities, relationships, and patterns) formulate

体现方式

学生需要认识到：

这是一个**线性关系问题**：

- 成本 = 固定成本 + 单位成本 × 数量

结构形式：

$$C = a + bx$$

这是现实中的**成本函数模型。**

6️⃣ Identify constraints, assumptions and simplifications

对应条目：

12. Identify the constraints, assumptions simplifications in a mathematical model formulate

体现方式

隐含假设包括：

- 每张 DVD 价格固定
- 不考虑折扣
- DVD 数量必须是整数
- 会员费一年只付一次

这些都是建模中的**简化假设。**